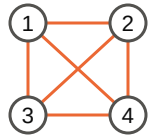
# Four boards, one token stream

Each board holds a quarter of every weight matrix; all four work on the same token at the same time

**PC on the LAN**
starts the four ranks over SSH and sends prompts to board 1's OpenAI-compatible API

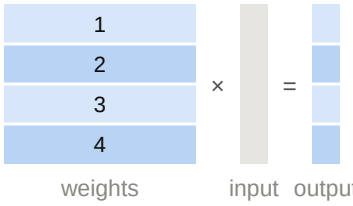
**Gigabit Ethernet switch**
jumbo frames (MTU 9000)



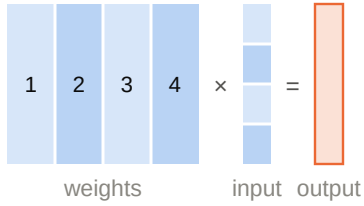

**Between the boards:**
a TCP full mesh, one connection per pair (6 in all)

| Board 1 | Board 2 | Board 3 | Board 4 |
|---|---|---|---|
| rank 0 | rank 1 | rank 2 | rank 3 |
| kv260-01 | kv260-02 | kv260-03 | kv260-04 |
| **CPU: one A53 core** attention for query heads 0-7; norms, sampling | **CPU: one A53 core** attention for query heads 8-15; norms, sampling | **CPU: one A53 core** attention for query heads 16-23; norms, sampling | **CPU: one A53 core** attention for query heads 24-31; norms, sampling |
| **FPGA engine + DDR4** its quarter of every matrix: 173.8 MB of 4-bit weights | **FPGA engine + DDR4** its quarter of every matrix: 173.8 MB of 4-bit weights | **FPGA engine + DDR4** its quarter of every matrix: 173.8 MB of 4-bit weights | **FPGA engine + DDR4** its quarter of every matrix: 173.8 MB of 4-bit weights |
| **serves the API (HTTP)** | gets API prompts from board 1 | gets API prompts from board 1 | gets API prompts from board 1 |

## How a weight matrix is split across the four boards




**By output rows**
q, k, v, gate, up and the output head (lm_head): each board computes its own quarter of the outputs, with no all-reduce




**By input columns**
attention output, FFN down: each board has a partial sum of every output; an all-reduce adds the four in board order, the same way on every board

## One token

each of the 16 layers: q, k, v → attention → attention output → all-reduce → gate, up → FFN down → all-reduce

then once per token: output head (lm_head) → argmax exchange → next token, the same on all four boards

33 exchanges per token: 2 all-reduces in each of the 16 layers, plus 1 argmax exchange.

engine (FPGA) exchange (network) CPU (one A53 core)

Llama-3.2-1B-Instruct, every weight matrix ggml Q4_0: 695.1 MB of weights per token, 173.8 MB per board on four.
Runtime: board/tp/tp_llama.cpp. Partial sums are added in rank order on every board, so all four hold the same hidden state.